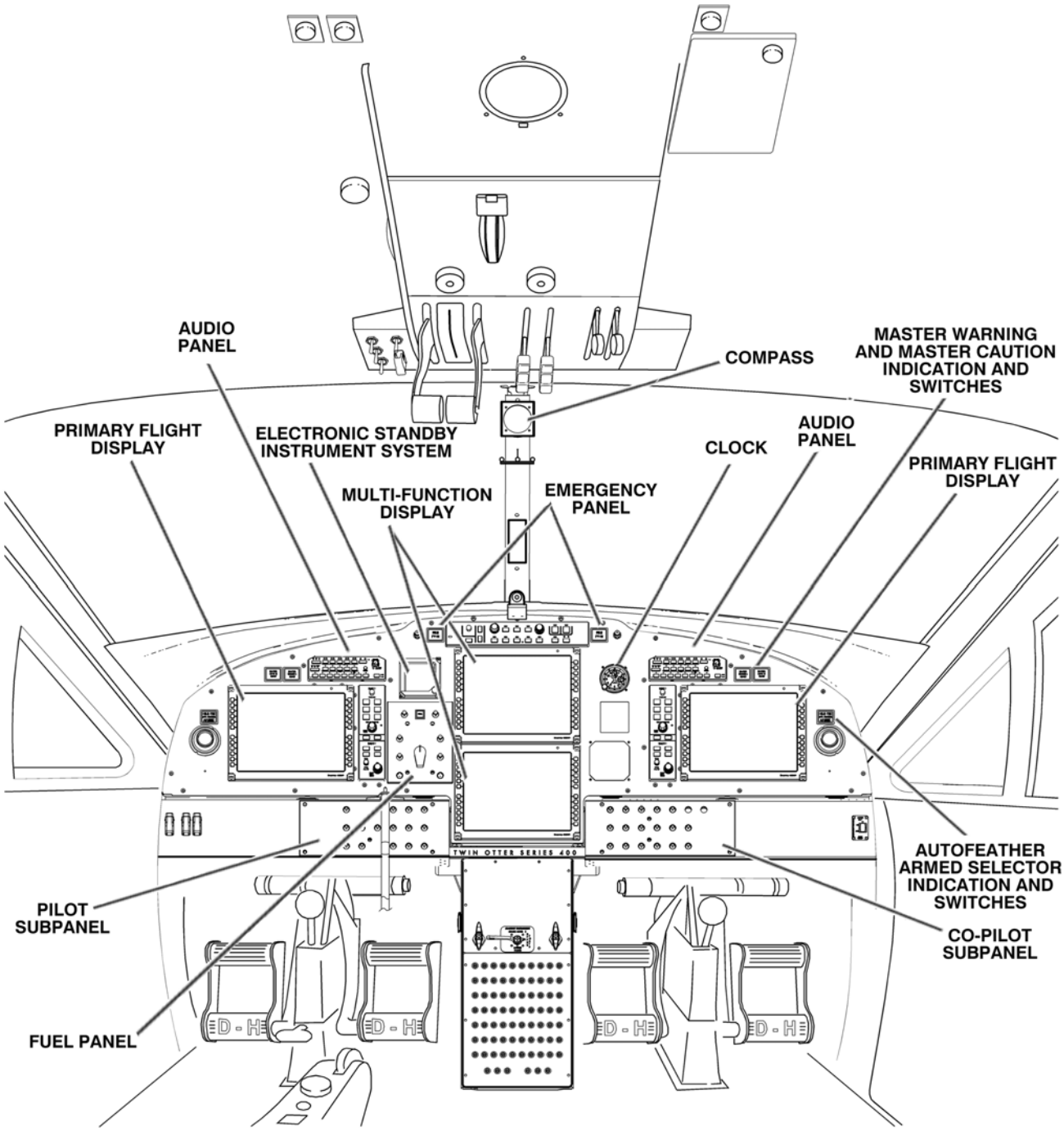
# Instrument and Control Panels – General Data

1. General

Refer to Figure 1 for details.

The instrument and control panels are comprised of the Pilot's and Co-pilot's Primary Flight Displays, the Electronic Standby Instrument System Display, the fuel panel, stall caution light, clock, master warning/caution indication, the propeller autofeather control switch and indication, Pilot's and Co-pilot's sub panels, the emergency panel, the upper and lower Multi-Function Displays.

The instrument panel is located across the front of the flight compartment below the windshield and secured to the A frame which is shock mounted to the aircraft. The area between the multi-function display panels and the Co-pilot's Primary Flight panel is reserved for customer standard option radio equipment. Refer to 23-00-00, Communications – General Data.




Instrument Panel
Figure 1

2. Description

A. Primary Flight Display (PFD)

The Pilot's and Co-pilot's Primary Flight Display shows all of the essential flight and navigation information, the engine instruments and radio tuning settings. Each Primary Flight Display screen presents, under normal operations, four predefined windows.

The Flight Display Window is comprised of the following displays:

- Attitude Director Indicator (ADI) Display
- Horizontal Situation Indicator (HSI) Display
- Engine Instruments Display
- Radio Tuning Display

28 Volt DC power is protected by a 5 Amp PILOT PFD (H1) (Left Bus) and COPILOT PFD (H2) (Right Bus) circuit breaker, located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

1) Attitude Director Indicator (ADI) Display

Refer to Figure 2 for details.

The Attitude Director Indicator (ADI) Display contains the essential flight information such as aircraft Pitch/Roll Attitude, Slip/Skid, Airspeed, Altitude, Vertical speed, Altimeter barometric setting, Rate of Turn indication, Composite lateral and vertical deviations.

Airspeed information displayed is:

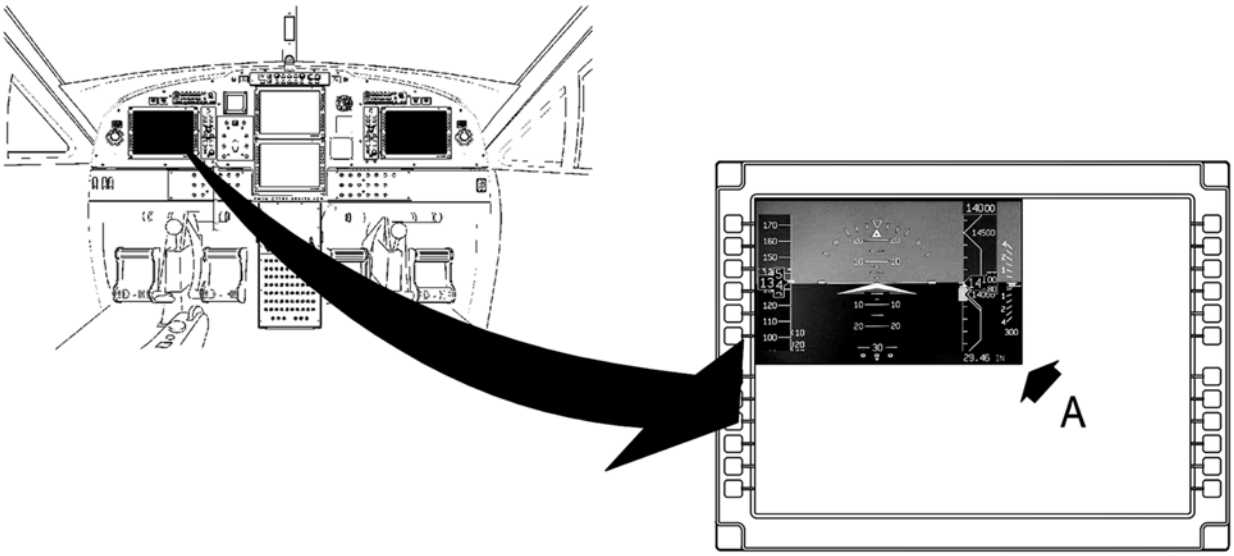
- Selected speed reference digital readout and bug
- Indicated airspeed tape and rolling digits
- Airspeed trend vector
- Maximum Operating Speed Indication
- $V_{SPEED}$ bugs
- Flaps and gear speed bugs
- Tape markers for $V_{YSE}$ and $V_{MC}$
- Low speed awareness bar and Stall annunciation
- Airspeed target bug

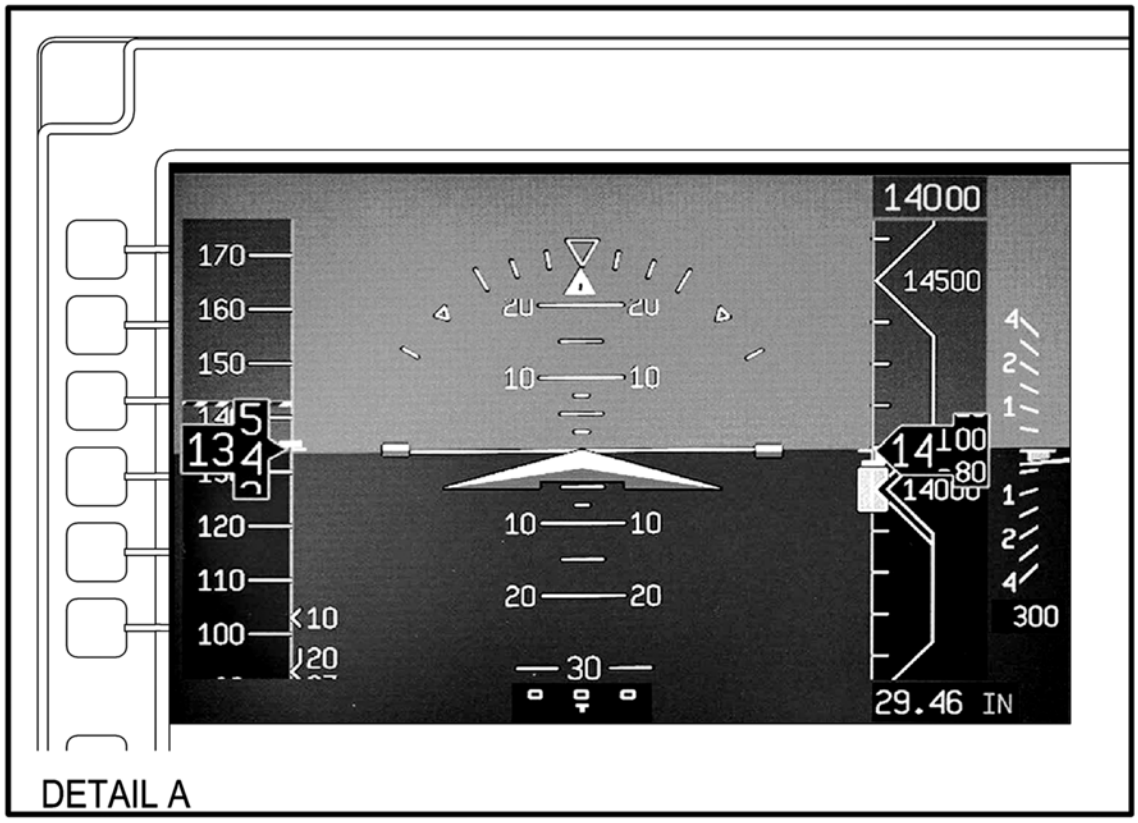
Altitude information displayed is:

- Altitude pitch tape
- Altitude rolling digits
- Metric altitude (when selected)
- Altitude trend vector
- Altitude selection and bug
- Radio attitude digital readout
- Barometric Altimeter Correction
- Minimum descent attitude and bug

Vertical Speed information displayed is:

- Vertical Speed Scale and pointer
- Vertical Speed digital readout
- Vertical Speed Target and bug




Attitude Director Indicator (Pre Mod 6/2303)
Figure 2

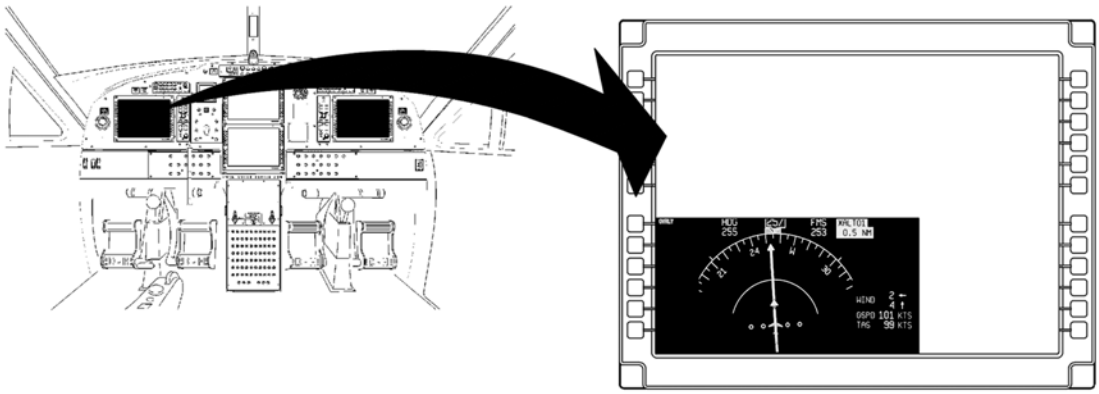
2) Horizontal Situation Indicator (HSI) Display

Refer to Figure 3 for details.

The Horizontal Situation Indicator (HSI) Display contains the navigational information required by the pilot or co-pilot. The information displayed is wind speed, ground and true airspeed, full and partial compass heading with selected course track and digital readout, lateral deviation indication and navigation system generated annunciations, refer to Monitor Warning System, 31-50-00, Monitor Warning System – General Data. Information also displayed as an HSI overlay includes TAWS generated alerts, TCAS, and Weather Radar. The navigational information is generated from several sources; the VOR/LOC, ADF, or FMS.

Information that may be overlaid on the HSI includes:

- Weather Overlay
- Terrain Awareness Display Overlay
- LX (lightening) Overlay
- TCAS



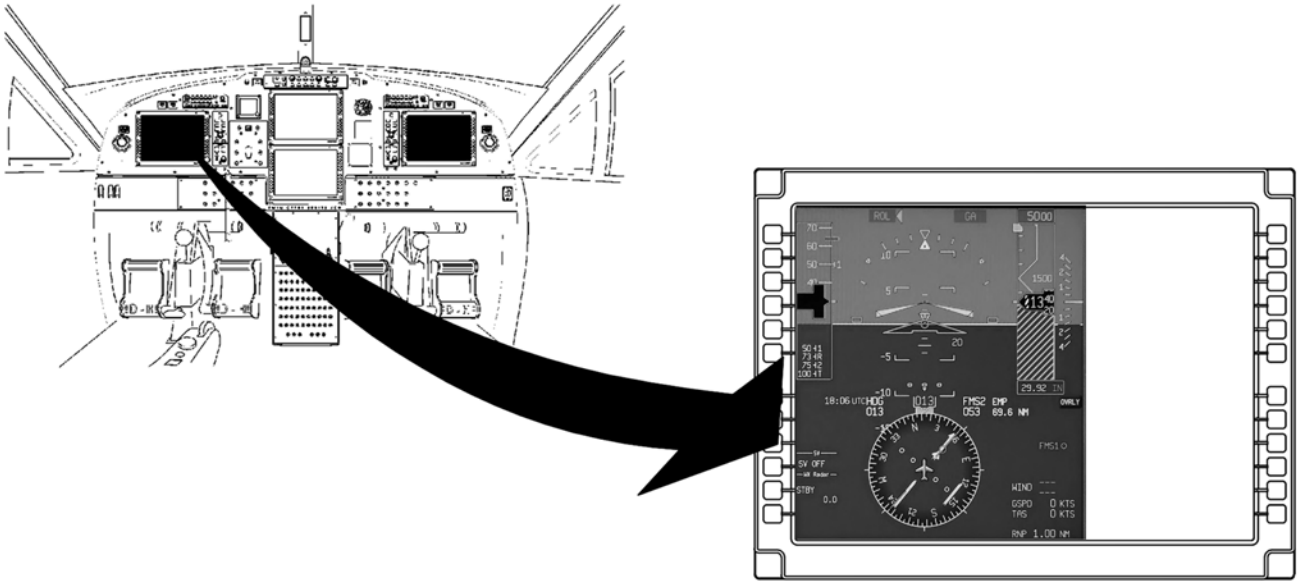



Horizontal Situation Indicator (Pre Mod 6/2303)
Figure 3

3) Integrated Primary Flight Display (IPFD) (Mod 6/2303)

Refer to Figure 4 for details.

Aircraft incorporating Mod 6/2303, have an Integrated Primary Flight Display (IPFD) which enhances the images presented on both PFD's, to provide a more representative aspect ratio from the pilot's eye point blue over brown representation. The IPFD combines traditional ADI and HSI displays into one window, allowing the attitude display to continue behind the HSI. The aspect ratio of the IPFD presents a pitch range of +15 to –20 degrees, relative to the aircraft's current attitude.



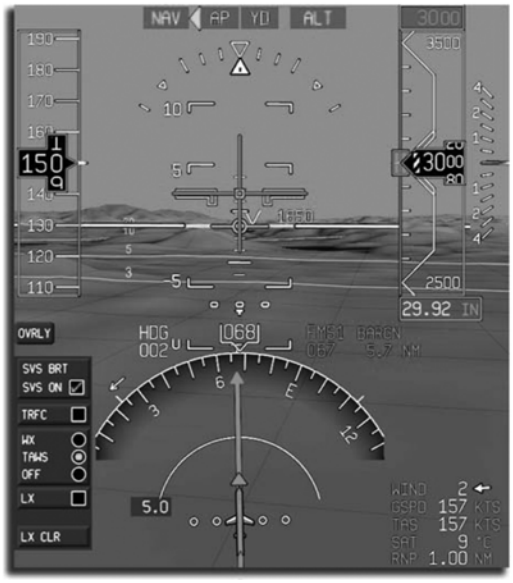
Attitude Director and Horizontal Situation Indicators (Post Mod 6/2303)
Figure 4

4) SmartView Synthetic Vision System (S.O.O. 6222)

Refer to Figure 5 for details.

Provisions have been made for SmartView Synthetic Vision System (SVS) to be enabled through the customer standard option S.O.O. 6222.

The SVS is automatically activated and displayed when the aircraft is powered on. The SVS is turned on and off by selecting or deselecting the SVS ON checkbox from the OVRLY menu. When the SVS ON checkbox is deselected, the standard IPFD standard blue over brown representation replaces the terrain on the PFD. SVS provides a computer generated three dimensional view of the terrain forward and below the aircraft's flight path, obstacles and runways, in addition to standard flight and navigation data. Aspect ratio of the SVS display matches that of the IPFD and more correctly represents the view of the outside world to enhance pilot situational awareness.




SmartView Synthetic Vision System (SVS) (S.O.O. 6222)
Figure 5

5) Engine Instrument Window

Refer to Figure 6 for details.

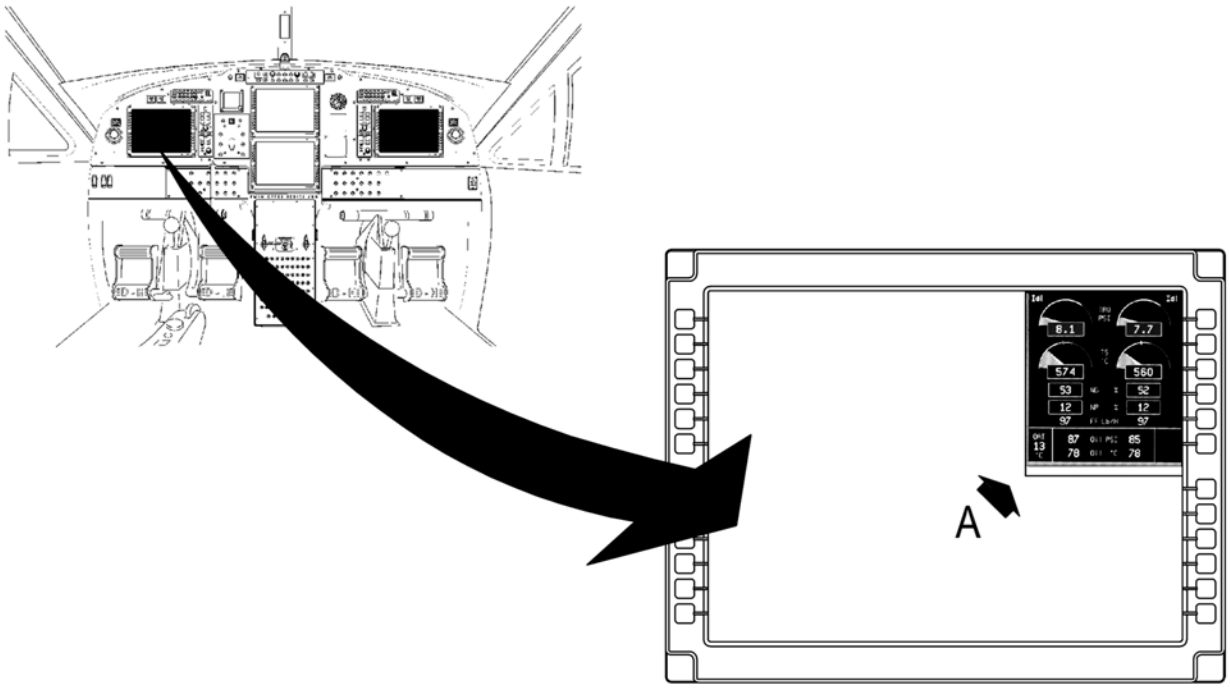
The engine instrument displays two sets of engine operating data, one for each engine.

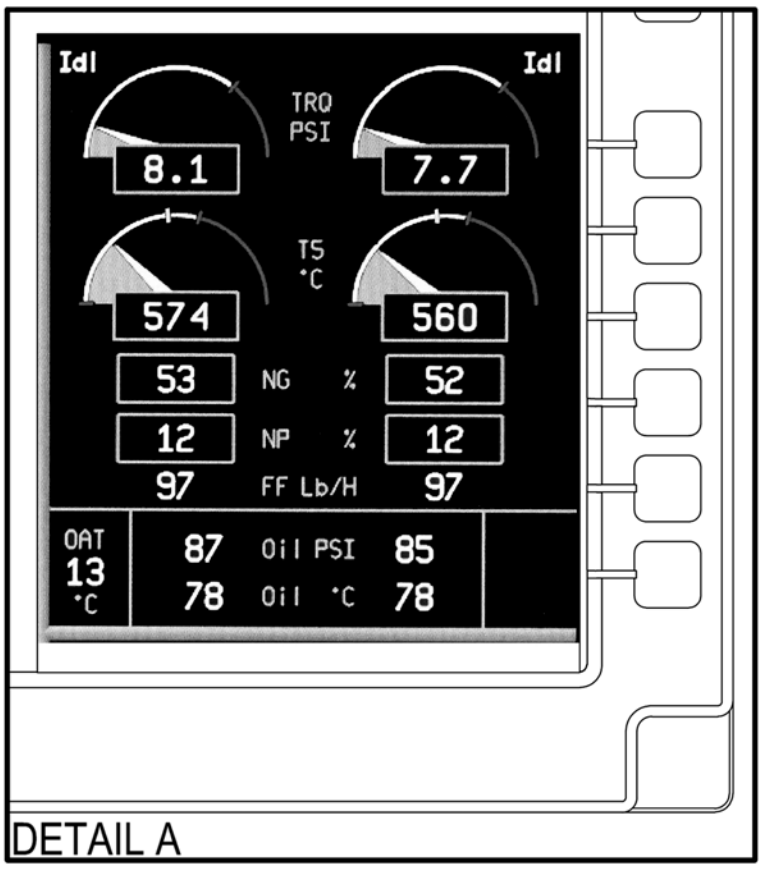
The engine instrument window information displayed is:

- Torque pressure indication
- Propeller tachometer indication
- Gas generator indication
- Fuel flow indication
- Turbine temperature indication
- Oil pressure indication
- Oil temperature indication
- Intake deflector indication
- Outside air temperature
- Engine Limits annunciation
- Start annunciation
- Anti-Ice annunciation (if installed)
- Propeller beta range annunciation
- Ignition annunciation

The torque pressure indication and the engine temperature indication are depicted by a pointer type display and integer values.

For aircraft incorporating S.O.O. 6280, FAA Engine Display Colours, the normal engine operating range parameters are displayed with a green arc or green line.



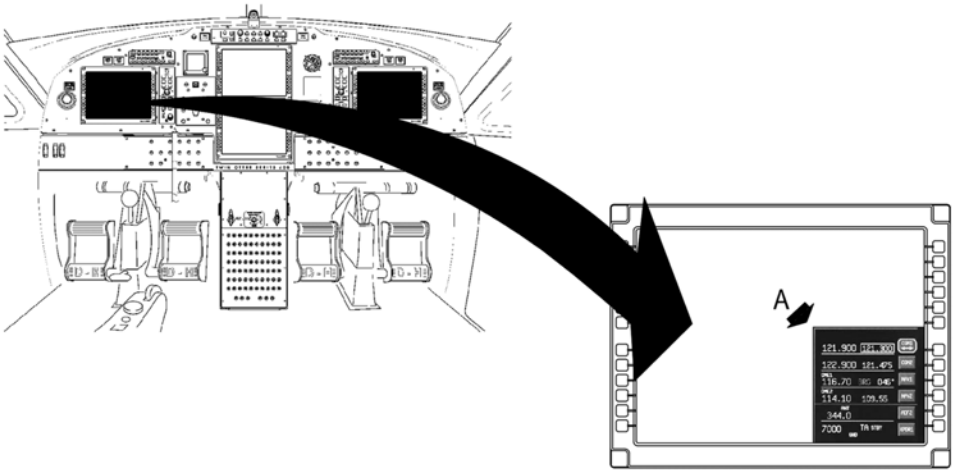

Engine Instrument Window
Figure 6

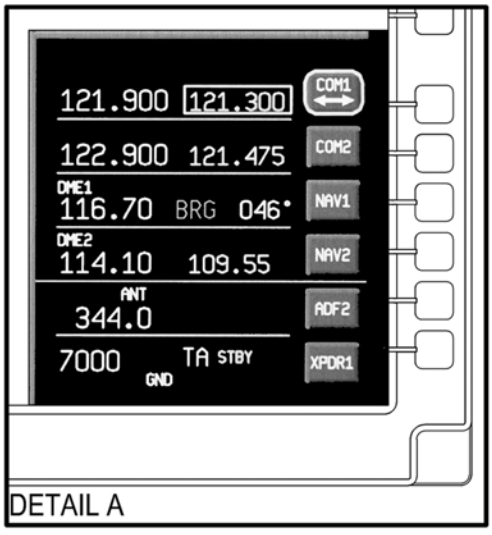
6) Radio Tuning Window

Refer to Figure 7 for details.

The radio tuning window is an interactive window which displays the selected operation frequencies for the following systems:

- COM 1
- COM 2
- NAV 1
- NAV 2
- Selected ADF
- Selected XPDR



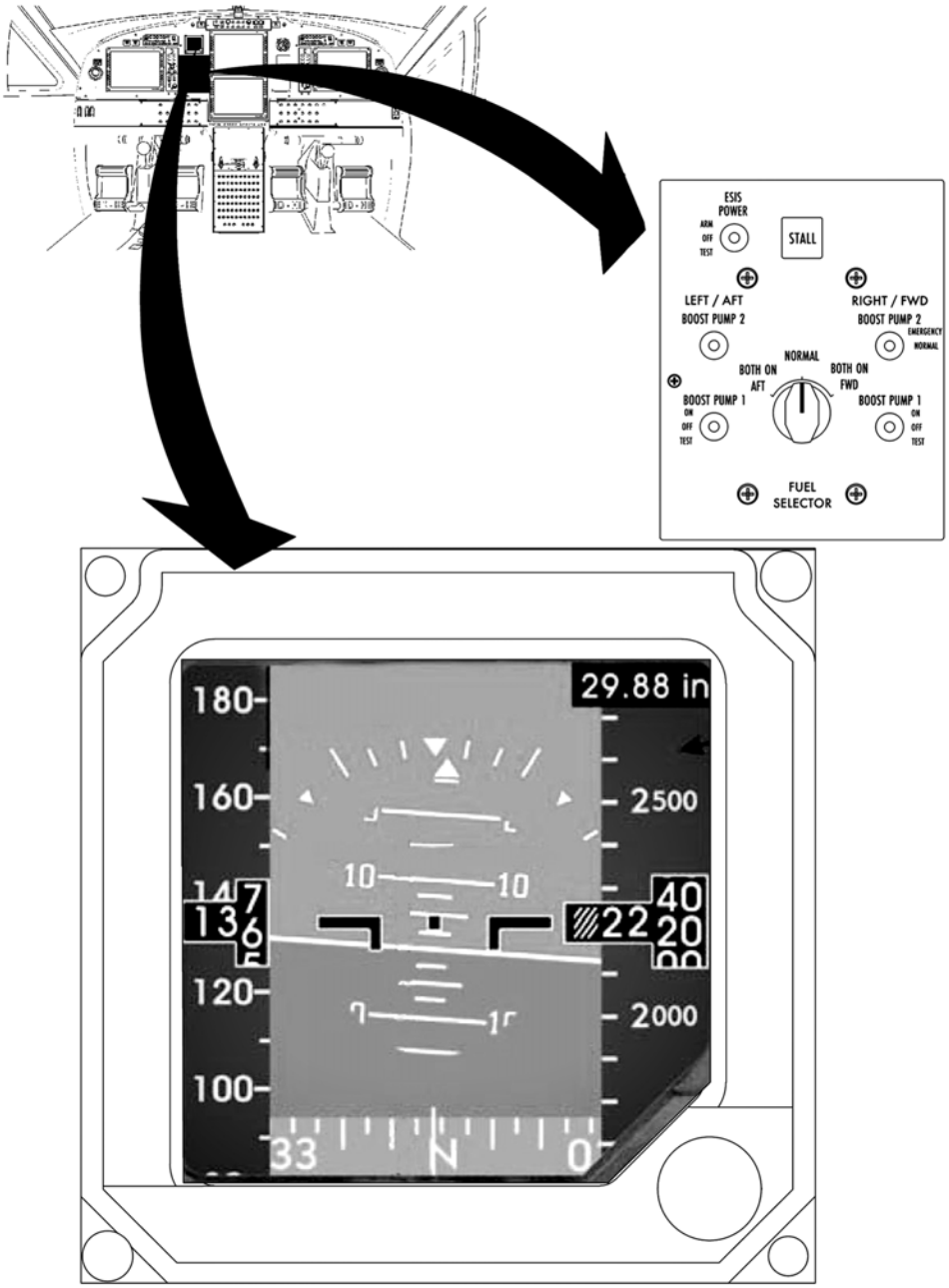




Radio Tuning Window
Figure 7

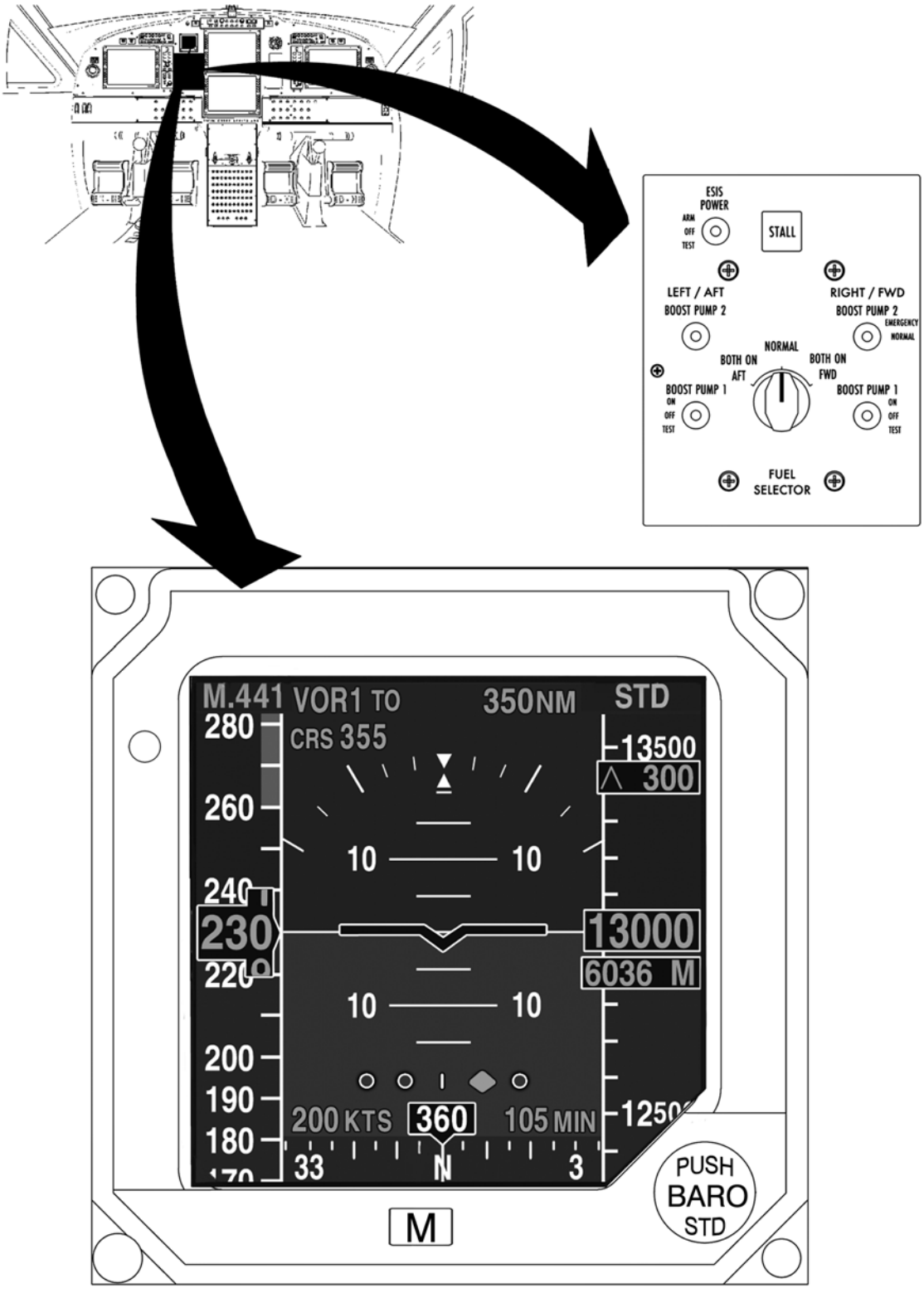
B. Electronic Standby Instrument System (ESIS)

Refer to Figure 8 and Figure 9 for details.

The ESIS instrument is powered by its own dedicated backup battery, and is entirely independent of the Apex system. When fully charged, the backup battery has sufficient capacity to operate the ESIS for at least 6 hours. Refer to 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049 and Mod 6/2170) – General Data.




L3 GH-3100 Electronic Standby Instrument System (Pre Mod 6/2170)
Figure 8




L3 GH-3900.2 Electronic Standby Instrument System (Post Mod 6/2170)
Figure 9

C. Multi-Function Display (MFD)

1) The upper multi-function display, presents the situational awareness information, under normal operation.

The display is divided into 3 predefined windows:

- Way point list
- Flight plan data
- Flight plan map

Provisions have been made for Vertical Situation Display (VSD) to be enabled through the incorporation of customer standard option S.O.O. 6312. When selected, VSD depicts a profile view of the aircraft's predicted flight path in relation to the terrain along the planned route or track in the lower 1/4 of the INAV Map window. VSD function is selected via the soft key on the Map window. The VSD selection is saved (retained) through an aircraft power cycle.

28 Volt DC power is protected by a 5 Amp UPPER MFD (J1) (Left Bus) and LOWER MFD (J2) (Right Bus) circuit breaker, located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

2) The lower multi-function display presents the system information under Normal operation, divided into predefined windows:

a) Crew Alerting System Window

The Crew Alerting System window displays the visual system warnings that are generated from system malfunctions or abnormal operating conditions. The visual warning may be accompanied by an aural warning. Refer to 31-50-00, Monitor Warning System – General Data.

b) Flight Control and Hydraulic Window

Flight Control and Hydraulic Window displays:

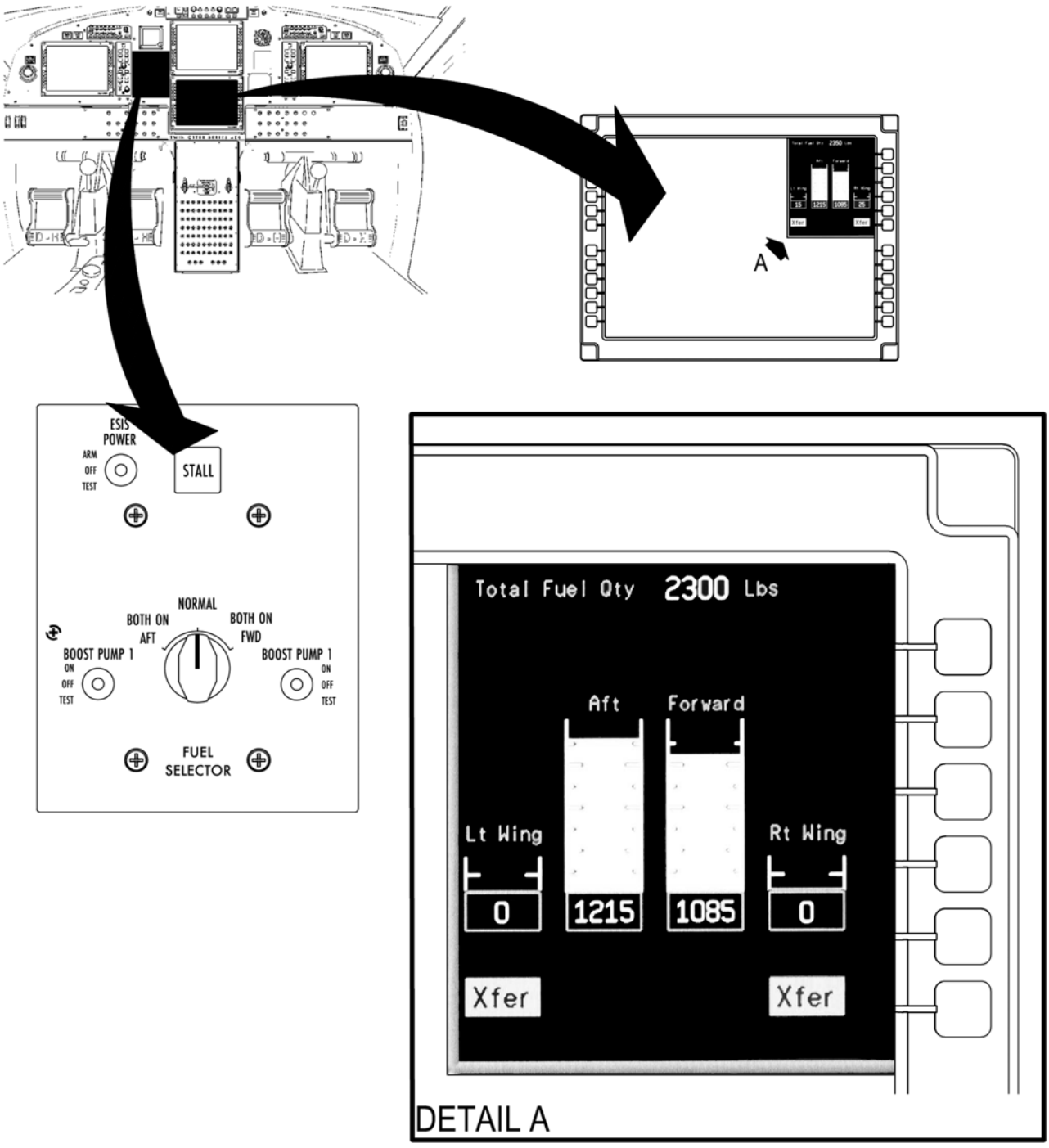
- Flap position Indication
- Rudder Trim Indication
- Aileron Trim Indication
- Pitch Trim Indication
- Hydraulic Pressure Indication
- Brake Pressure Indication

c) Fuel Quantity Window

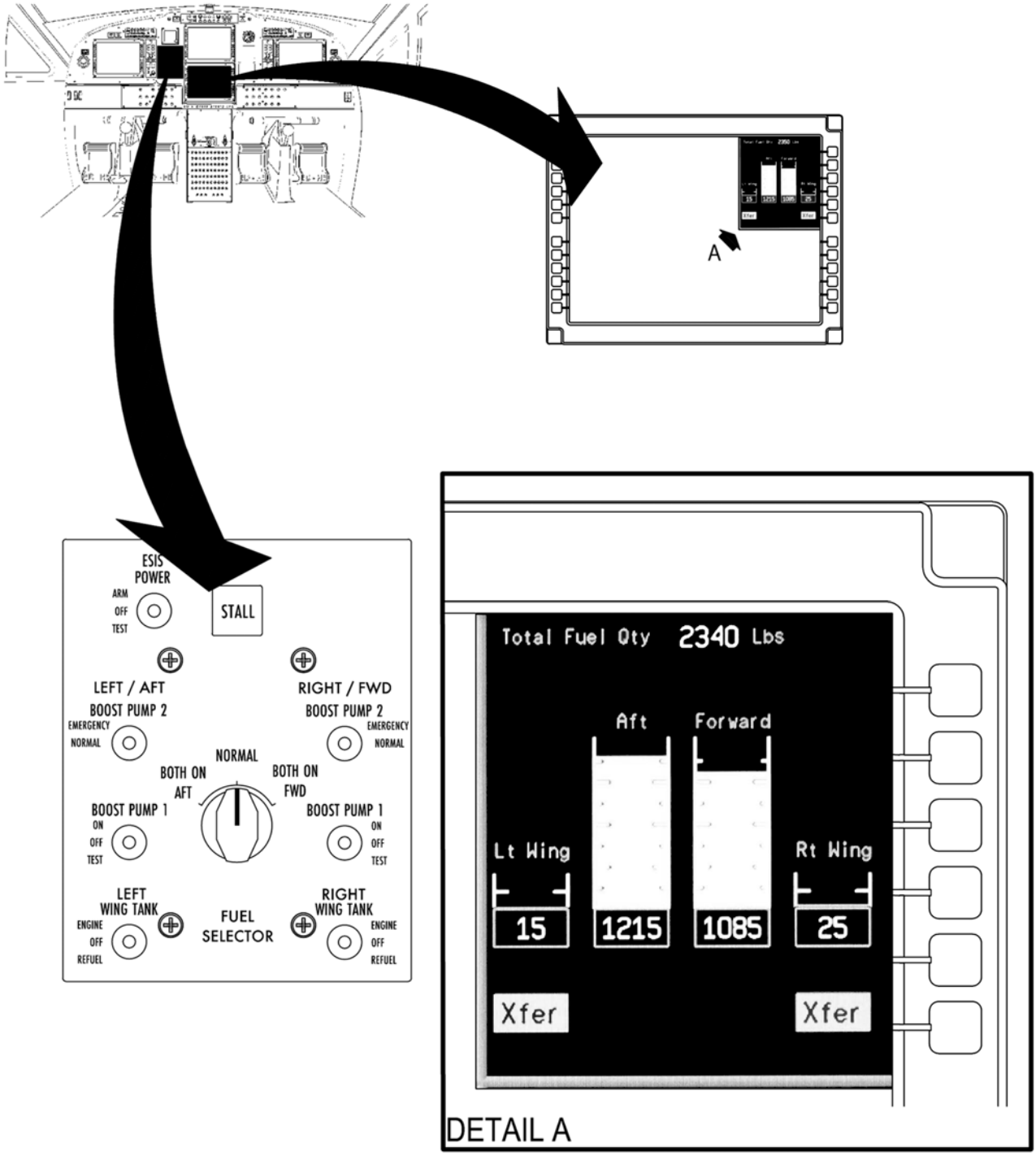
Refer to Figure 10 for details.

Fuel Quantity Window displays:

- Total Fuel Quantity in LBS
- Cross Feed Valve Status
- Fuel Tank Level Status for the Aft, Forward, Left and Right wing, (if installed) Tank status
- Boost Pump Status
- Wing Pump Status (if installed) (refer to Figure 11)




Fuel Quantity Window
Figure 10




Wing Pump Status (S.O.O. 6247)
Figure 11

d) Environmental Display

Environmental Window displays:

- Cockpit temperature
- Cabin temperature

e) Miscellaneous Window

Miscellaneous Window displays:

- Electrical System Indication
- Ice Protection status

f) Electronic Checklist Window

The electronic checklist window displays the predefined checklist but also can display the following windows:

- Navigation Sensor Window
- Weather/Lightning/Terrain Awareness and Warning System Window
- Avionics Window
- Software configuration and data loading window

D. Fuel Panel

The fuel panel contains the fuel control switches, the standby booster pump switches and the Stall Warning Light. The panel is internally lit with LED lights. (Refer to 33-00-00, Lights – General Data).

E. Clock

The clock is mounted on the instrument panel right of the multi-function displays. Refer to 31-20-11, Clock (Mod 6/2049) – General Data.

F. Master Warning and Master Caution Indication and Switches

Two 1 inch by 1.2 inch LED switchlights, internally labelled MASTER CAUTION and MASTER WARNING, are provided directly above each pilot's primary flight display, refer to 31-50-00, Monitor Warning System – General Data.

G. Autofeather Switch/Indicators

Two identical propeller autofeather switches are provided, located outboard of each pilot's primary flight display. When activated (accomplished by pressing either switch, selected autofeather armed) electrical power is supplied to the autofeather system as indicated by the illumination of SELECT on two switches, refer to 61-22-02, Propeller Autofeather System (Mod 6/2022) – General Data.

H. Pilot's Subpanel

The Pilot's Subpanel contains the System Test switches, the control switches for the Ice & Rain, External Lighting and Internal Lighting. The panel is backlit with LED lights. (Refer to 33-00-00, Lights – General Data).

I. Co-pilot's Subpanel

The Co-pilot's Subpanel contains the control switches for Environmental, Surface De-Ice, Emergency and the Artex ELT Arming switch. The panel is backlit with LED lights. (Refer to 33-00-00, Lights – General Data).

J. Emergency Panel

The emergency panel is a designated area at the top of the instrument panel where the left and right fuel emergency shut-off switches, the integral fire warning light/extinguisher push buttons are located. (Refer to 26-10-00, Fire Detecting System – General Data).

3. Removal/Installation

A. Removal of Flight Instrument Panel

1) Ensure all aircraft electrical power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Remove all instruments. (Refer to Instrument Chapters).

3) Disconnect electrical connectors, bonding wire and pressure lines. Cap connectors.

4) Support panel and remove mounting screws.

5) Remove panel from aircraft.

B. Installation of Flight Instrument Panel

1) Install panel in aircraft.

2) Support panel and install mounting screws.

3) Uncap and connect electrical connectors, bonding wire and pressure lines.

4) Install all instruments. (Refer to Instrument Chapters).

C. Removal of the Lower Shockmounts

1) Remove instrument panel, refer to Para 3.A.

2) Remove rivets securing shockmount to structure by carefully drilling out manufactured rivet head and tap out shank with suitable punch.

D. Installation of the Lower Shockmount

1) Position shockmount on structure mounting point as the removed one and secure with four rivets (MS20470–AD5).

2) Install instrument panel, refer to Para 3.B.